

Differential Equations and Applications



Lecture 2: MATH 203i

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Equations reduce to separable

The D.E. y' (or x') $= f(ax + by + c)$ can be reduced to separable by substitution $u = ax + by + c \rightarrow du = adx + bdy$

Example: Solve the D.E. $y' = \tan^2(x + y + 2)$

Solution : Let $u = x + y + 2 \rightarrow du = dx + dy \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$

$$\frac{du}{dx} - 1 = \tan^2 u$$

$$\frac{du}{dx} = 1 + \tan^2 u = \sec^2 u \text{ separable D.E.}$$

$$\cos^2 u \, du = dx \quad \text{integrating both sides}$$

$$\int \frac{1}{2} + \frac{1}{2} \cos 2u \, du = x + C$$

$$\frac{u}{2} + \frac{1}{4} \sin 2u = x + C \rightarrow \frac{(x + y + 2)}{2} + \frac{1}{4} \sin 2(x + y + 2) = x + C$$

Example:- Solve the following DEs

$$y' = (x + y + 1)^2$$

$$\text{Let } z = (x + y + 1) \Rightarrow z' = 1 + y' \Rightarrow y' = z' - 1$$

$$z' - 1 = (z)^2$$

$$z' = 1 + z^2 = \frac{dz}{dx}$$

$$\frac{dz}{1 + z^2} = dx$$

$$\tan^{-1} z = x + c$$

$$z = \tan(x + c)$$

$$\text{let } z = x + y + 1$$

$$y = \tan(x + c) - x - 1$$

Example:- Solve the following DEs

$$(x - y)dx + (y - x - 2)dy = 0, \quad y(0) = 2$$

$$y' = \frac{x - y}{x - y + 2}$$

$$\text{Let } z = (x - y) \rightarrow z' = 1 - y' \rightarrow y' = 1 - z'$$

Substitute in the DE

$$y' = \frac{x - y}{x - y + 2} \rightarrow 1 - z' = \frac{z}{z + 2}$$

$$1 - z' = \frac{z}{z + 2}$$

$$\frac{dz}{dx} = 1 - \frac{z}{z + 2} = \frac{2}{z + 2}$$

$$\int (z + 2)dz = \int 2dx \rightarrow z^2 + 4z = 4x + c$$

$$(x - y)^2 + 4(x - y) = 4x + c, \quad y(0) = 2 \rightarrow c = -4$$

Example:- Solve the following DEs

$$3y' + 2 = e^{-(2x+3y)} / (2x + 3y)$$

Let $z = (2x + 3y) \Rightarrow z' = 2 + 3y'$

$$z' = e^{-z} / z$$

$$\frac{dz}{dx} = \frac{1}{ze^z}$$

$$\int ze^z dz = \int dx$$

$$ze^z - e^z = x + c$$

$$e^{(2x+3y)}(2x + 3y - 1) = x + c$$

Homogeneous Differential Equation

Homogenous function

The function $f(x, y)$ is called homogeneous of degree n if

$$f(tx, ty) = t^n f(x, y)$$

Example: the function $f(x, y) = 2x^3 + 5y^3 + x^2y$

is a homogeneous function of degree 3, since

$$\begin{aligned} f(tx, ty) &= 2(tx)^3 + 5(ty)^3 + (tx)^2(ty) = t^3(2x^3 + 5y^3 + x^2y) \\ &= t^3 f(x, y). \end{aligned}$$

Whereas $f(x, y) = x^3 + y^3 + 1$ is not homogeneous.

Homogeneous Differential Equation

The D.E. $M(x, y)dx + N(x, y)dy = 0$

is called homogeneous if both M, N are homogeneous functions of the same degree.

Solving homogeneous D.E.

The homogeneous D.E. can be transformed to separable D.E. using one substitution of the following:

$$1. \quad \frac{y}{x} = u \rightarrow y = xu \rightarrow \frac{dy}{dx} = x \frac{du}{dx} + u$$

$$2. \quad \frac{x}{y} = v \rightarrow x = yv \rightarrow \frac{dx}{dy} = y \frac{dv}{dy} + v$$

Then solving the separable D.E. between (u, x) or (v, y) and finally back substituting u or v in terms of x and y

Example

Solve the D.E. $(y + \sqrt{x^2 + y^2}) dx - x dy = 0$

Solution: The D.E. is homogeneous since M , N are homogeneous functions of degree 1

$$\text{Let } \frac{y}{x} = u \rightarrow y = xu \rightarrow \frac{dy}{dx} = u + x \frac{du}{dx}$$

$$(ux + \sqrt{x^2 + u^2 x^2}) - x \left(u + x \frac{du}{dx} \right) = 0 \text{ divide by } x \text{ and simplify}$$

$$(\cancel{u} + \sqrt{1 + u^2}) - (\cancel{u} + x \frac{du}{dx}) = 0$$

$$\sqrt{1 + u^2} - x \frac{du}{dx} = 0 \rightarrow \frac{du}{\sqrt{1 + u^2}} = \frac{dx}{x}$$

$$\sinh^{-1} u = \ln|x| + C$$

$$\sinh^{-1} \frac{y}{x} = \ln|x| + C$$

Example 2

Solve the IVP:

$$\frac{dy}{dx} = \frac{y}{x} (1 + \ln y - \ln x) \quad \text{and} \quad y(1) = e.$$

Solution :

$$\frac{dy}{dx} = \frac{y}{x} \left(1 + \ln \frac{y}{x} \right)$$

$$\text{Let } u = \frac{y}{x} \quad \text{i.e.} \quad y = x u \quad \text{and} \quad \frac{dy}{dx} = x \frac{du}{dx} + u$$

$$x \frac{du}{dx} + u = u(1 + \ln u) = u + u \ln u$$

$$x \frac{du}{dx} = u \ln u \quad \Rightarrow \quad \int \frac{du}{u \ln u} = \int \frac{dx}{x} \quad \Rightarrow \quad \ln |\ln u| = \ln |x| + c$$

$$\ln \left| \ln \left(\frac{y}{x} \right) \right| = \ln |x| + c$$

$$y(1) = e \quad \Rightarrow \quad \ln |\ln e| = \ln 1 + c \quad \Rightarrow \quad c = 0$$

Equations reduce to homogeneous

The D.E. on the form $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$

Can be reduced to homogeneous or separable depending on a_1, b_1, a_2 and b_2

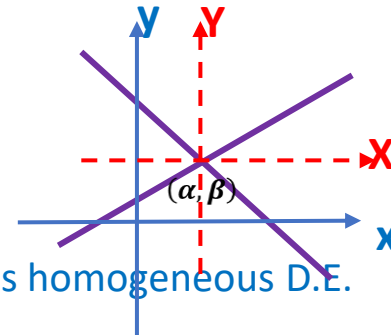
Case 1: If $\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$

lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are intersecting at (α, β)

Transform axes to (α, β) i.e. let $x = X + \alpha$, $y = Y + \beta$

New equation: $(a_1X + b_1Y)dX + (a_2X + b_2Y)dY = 0$ which is homogeneous D.E.

After solving the homogeneous D.E. we substitute back $X = x - \alpha$, $Y = y - \beta$
to find the solution of the original problem in x, y



Example

Solve the D.E. $(x - 2y + 1)dx + (4x - 3y - 6)dy = 0$

Solution: $\frac{a_1}{b_1} = -\frac{1}{2}, \frac{a_2}{b_2} = -\frac{4}{3} \neq \frac{a_1}{b_1}$ (case 1)

Solving $x - 2y + 1 = 0, 4x - 3y - 6 = 0 \rightarrow (\alpha, \beta) = (3, 2)$

Let $x = X + 3, y = Y + 2$

$(X - 2Y)dX + (4X - 3Y)dY = 0$ which is homogeneous

Let $\frac{Y}{X} = u \rightarrow Y = Xu \rightarrow dY = Xdu + udX$

$(X - 2uX)dX + (4X - 3uX)(Xdu + udX) = 0$ divide by X

$(1 - 2u)dX + (4 - 3u)(Xdu + udX) = 0$

$(1 - 2u + 4u - 3u^2)dX + (4 - 3u)Xdu = 0 \rightarrow \text{separable D.E.}$

$(1 + 2u - 3u^2)dX + (4 - 3u)Xdu = 0$

Example(cont.)

$(3u^2 - 2u - 1)dX + (3u - 4)Xdu = 0$ separating variables

$$\frac{-dX}{X} = \frac{(3u-4)du}{(3u^2-2u-1)} \quad \text{Integrating both sides}$$

$$\int \frac{-dX}{X} = \int \frac{(3u-4)du}{(3u^2-2u-1)} = \int \frac{(3u-4)du}{(3u+1)(u-1)} \quad \text{using partial fractions}$$

$$-\ln X + \ln c = \int \frac{\frac{15}{4}}{(3u+1)} - \frac{\frac{1}{4}}{u-1} du = \frac{5}{4} \ln(3u+1) - \frac{1}{4} \ln(u-1)$$

$$4 \ln \left(\frac{c}{X} \right) = 5 \ln(3u+1) - \ln(u-1) = \ln \frac{(3u+1)^5}{u-1}$$

$$\frac{c}{X^4} = \frac{(3\frac{Y}{X}+1)^5}{\frac{Y}{X}-1} \rightarrow \frac{c}{(x-3)^4} = \frac{(3\frac{y-2}{x-3}+1)^5}{\frac{y-2}{x-3}-1}$$

Case 2: If $\frac{a_1}{b_1} = \frac{a_2}{b_2}$

the D.E. can be reduced to separable by substitution :

$$z = a_1x + b_1y \rightarrow dz = a_1dx + b_1dy$$

Example

Solve the D.E. $(x + 2y + 3)dx + (2x + 4y - 1)dy = 0$

Solution: $\frac{a_1}{b_1} = \frac{1}{2}, \frac{a_2}{b_2} = \frac{1}{2} = \frac{a_1}{b_1}$ (case 2)

Let $z = x + 2y \rightarrow dz = dx + 2dy$

$$(z + 3)dx + (2z - 1)\left(\frac{dz - dx}{2}\right) = 0$$

$$(2z + 6)dx + (2z - 1)dz - (2z - 1)dx = 0$$

$$7dx + (2z - 1)dz = 0 \quad (\text{separable}) \rightarrow 7x + (z^2 - z) = C$$

$$7x + (x + 2y)^2 - (x + 2y) = C$$

Exact Differential Equations

The D.E. $M(x, y)dx + N(x, y)dy = 0$ is called exact if

$$\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x}$$

The solution function $F(x, y) = C$

The total differential of F is $dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0$ comparing this to the D.E. then it follows that

$$\frac{\partial F}{\partial x} = M(x, y) \quad \text{and} \quad \frac{\partial F}{\partial y} = N(x, y)$$

1. We can use the first equation to find the solution function F

$F = \int M(x, y)dx + g(y)$ then find $g(y)$ by substituting in the second equation $\frac{\partial F}{\partial y} = N(x, y)$.

Exact Differential Equations

2. We can use the second equation to find the solution function F

$F = \int N(x, y)dy + f(x)$ then find $f(x)$ by substituting in the first equation: $\frac{\partial F}{\partial x} = M(x, y)$

3. We can integrate the 2 equations then compare result of integration to find the solution function F

We will solve the following example using each the 3 alternatives for clarification.

Example

Solve the D.E. $(2xy - \sec^2 x)dx + (x^2 + 2y)dy = 0$

Solution: $\frac{\partial M}{\partial y} = 2x$ $\frac{\partial N}{\partial x} = 2x$ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow$ the D. E. is exact

Let $F(x, y) = C$ be the solution function

$$\frac{\partial F}{\partial x} = M = (2xy - \sec^2 x) \quad \frac{\partial F}{\partial y} = N = (x^2 + 2y)$$

First alternative: $F = \int (2xy - \sec^2 x)dx + g(y)$

$F = x^2y - \tan x + g(y)$ substitute in 2nd equation

$$\frac{\partial F}{\partial y} = x^2 + g'(y) = N = (x^2 + 2y) \rightarrow g'(y) = 2y \quad g(y) = y^2$$

$$x^2y - \tan x + y^2 = C$$

Example (Cont.)

Second alternative:

Start with $\frac{\partial F}{\partial y} = N = (x^2 + 2y) \rightarrow F = \int (x^2 + 2y) dy + f(x)$

$$F = x^2 y + y^2 + f(x) \quad \text{substitute in first equation}$$

$$\frac{\partial F}{\partial x} = 2xy + f'(x) = M = (2xy - \sec^2 x) \rightarrow f'(x) = -\sec^2 x$$

$$f(x) = -\tan x \quad \text{substitute in } F: \quad x^2 y + y^2 - \tan x = C$$

Third alternative:

$$\frac{\partial F}{\partial x} = M = (2xy - \sec^2 x) \rightarrow F = x^2 y - \tan x + g(y)$$

$$\frac{\partial F}{\partial y} = N = (x^2 + 2y) \rightarrow F = x^2 y + y^2 + f(x)$$

Comparing two expressions to find $f(x) = -\tan x$ and $g(y) = y^2 \dots$

Example 2

Solve the D.E. $(\cos x + \ln y) dx + \left(\frac{x}{y} + \frac{\sinh^{-1} y}{\sqrt{1+y^2}} \right) dy = 0$

Solution: $\frac{\partial M}{\partial y} = \frac{1}{y}$ $\frac{\partial N}{\partial x} = \frac{1}{y} = \frac{\partial M}{\partial y} \rightarrow D.E. \text{ is exact}$

Let the solution function $F(x, y) = C$

$$\frac{\partial F}{\partial x} = M = (\cos x + \ln y) \text{ and } \frac{\partial F}{\partial y} = N = \left(\frac{x}{y} + \frac{\sinh^{-1} y}{\sqrt{1+y^2}} \right)$$

$$F = \int (\cos x + \ln y) dx + g(y) = \sin x + x \ln y + g(y)$$

$$\frac{\partial F}{\partial y} = \frac{x}{y} + g'(y) = N = \left(\frac{x}{y} + \frac{\sinh^{-1} y}{\sqrt{1+y^2}} \right) \rightarrow g'(y) = \frac{\sinh^{-1} y}{\sqrt{1+y^2}}$$

$$g(y) = \frac{1}{2} (\sinh^{-1} y)^2 \rightarrow \text{solution: } \sin x + x \ln y + \frac{1}{2} (\sinh^{-1} y)^2 = C$$

End of Lect. 2